PROGRESSIVE ELECTRICITY BILLING SYSTEM

Python Programming Case Study

1. Problem Statement

Households use different amounts of electricity each month. Calculating bills manually across multiple consumption slabs, fixed charges, surcharges, and rebates can lead to mistakes. A program is needed to calculate monthly bills, compare bills across months, and identify unusually large increases.

2. Objective

To develop a Python program that calculates household electricity bills using progressive slabs, fixed charges, surcharges, and rebates, then compares monthly bills and flags unusual increases.

3. Assumptions

- The household enters electricity consumption in units (kWh) for each month.
- Energy charge slabs are progressive: first 100 units at ■1.50/unit; next 100 at ■2.50/unit; next 100 at ■4.00/unit; units above 300 at ■6.00/unit.
- A fixed charge of ■75 is added to every monthly bill.
- A surcharge of 5% is applied to the energy charge.
- A rebate of 5% of the energy charge is given when monthly consumption is 100 units or less.
- The final bill is energy charge + fixed charge + surcharge – rebate.
- An increase greater than 25% compared with the previous month's bill is flagged as unusual.

4. Algorithm

1. Start the program and read the number of months.
2. For each month, read the month name and electricity consumption in units.
3. Calculate energy charges progressively using the four consumption slabs.
4. Add the fixed charge and calculate the 5% surcharge.
5. If consumption is 100 units or less, calculate the 5% rebate on the energy charge; otherwise, set rebate to zero.
6. Calculate and store the final bill for the month.
7. Compare each month's bill with the previous month's bill and calculate the percentage increase.
8. If the increase is greater than 25%, mark it as an unusual increase.
9. Display monthly bill details, comparison results, and stop.

5. Flowchart

START
↓
Input number of months
↓
Read month name and units consumed
↓
Calculate energy charge using progressive slabs
↓
Add fixed charge and surcharge; check rebate
↓
Calculate and store final monthly bill
↓
More months? YES → Read next month
NO ↓
Display all monthly bills
↓
Compare bills with previous month
↓
Increase greater than 25%? → Flag unusual increase
↓
END

6. Pseudocode

```
BEGIN
    INPUT number_of_months
    bills = empty list

    FOR each month
        INPUT month_name, units
        energy = 0

        IF units <= 100
            energy = units * 1.50
        ELSE IF units <= 200
```

```
            energy = 100 * 1.50 + (units - 100) * 2.50
        ELSE IF units <= 300
            energy = 100 * 1.50 + 100 * 2.50
                     + (units - 200) * 4.00
        ELSE
            energy = 100 * 1.50 + 100 * 2.50
                     + 100 * 4.00 + (units - 300) * 6.00
        END IF

        fixed = 75
        surcharge = energy * 0.05

        IF units <= 100
            rebate = energy * 0.05
        ELSE
            rebate = 0
        END IF

        bill = energy + fixed + surcharge - rebate
        ADD month_name, units, energy, bill to bills
    END FOR

    DISPLAY each monthly bill
    COMPARE each bill with previous month's bill
    IF increase percentage > 25
        DISPLAY "Unusual increase"
    END IF
END
```

7. Python Code

```
bills = []
n = int(input("Enter number of months: "))

for i in range(n):
    print("\\nMonth", i + 1)
    month = input("Enter month name: ")
    units = float(input("Enter units consumed: "))

    if units <= 100:
        energy = units * 1.50
    elif units <= 200:
        energy = 100 * 1.50 + (units - 100) * 2.50
    elif units <= 300:
        energy = (100 * 1.50 + 100 * 2.50
                  + (units - 200) * 4.00)
    else:
        energy = (100 * 1.50 + 100 * 2.50
                  + 100 * 4.00 + (units - 300) * 6.00)

    fixed = 75
    surcharge = energy * 0.05

    if units <= 100:
        rebate = energy * 0.05
    else:
        rebate = 0

    bill = energy + fixed + surcharge - rebate
    bills.append([month, units, energy, fixed,
                  surcharge, rebate, bill])

print("\\nMONTHLY ELECTRICITY BILLS")
for item in bills:
    print(item[0], "Units:", item[1],
          "Energy:", round(item[2], 2),
          "Fixed:", item[3],
          "Surcharge:", round(item[4], 2),
          "Rebate:", round(item[5], 2),
          "Final Bill:", round(item[6], 2))

print("\\nMONTH-TO-MONTH COMPARISON")
for i in range(1, len(bills)):
    previous = bills[i - 1][6]
    current = bills[i][6]
    increase = ((current - previous) / previous) * 100

    print(bills[i][0], "Change:", round(increase, 2), "%")
    if increase > 25:
        print("Unusual increase detected")
    else:
        print("Normal change")
```

8. Expected Output (Illustration)

```
MONTHLY ELECTRICITY BILLS\nJanuary    Units: 80  Final Bill: INR 195.00\nFebruary   Units: 120 Final Bill: INR 285.00\nMarch      Units
```